

Smiling Back: Modeling and Evaluating Facial Reciprocity in Conversational Agents

Daniel Yang, Hirofumi Inaguma, Jason Zheng, Tu Anh Nguyen, Xutai Ma
USC Ming Hsieh Department of Electrical and Computer Engineering
Meta Fundamental AI Research



Motivation

The Problem: When Does Visual Perception Matter for Conversational Agents?

Talking Head Systems are Everywhere but...

- Wav2Lip, INFP, and recent diffusion models → **audio-only input**
- Listening head models → visually perceptive but **permanently mute** [Ng et al. 2022; Liu et al. 2023]
- Unified dyadic models → rarely target **sub-second visual reciprocity**

Timing is Everything (and Ignored):

- Natural responses must begin within **200-400ms** of perception
- Turn-taking gaps are only **~200ms median**
- Standard metrics (LPIPS, PSNR, FID) are **insensitive to timing**
- Models optimized for frame-quality become **one-way performers**, not partners

How do we build and evaluate agents that see, react, and reciprocate in real time?

Our Contribution: soft-FCCS and Benchmark

What Do We Measure?

- How well does an agent's facial dynamics **align temporally** with its partner within a short lag window?

FCCS (Evaluation): cross-correlation score between learned facial composites

$$\text{FCCS} = \max_{\tau \in [-L, L]} \text{corr}(\hat{y}_t, y_{t+\tau})$$

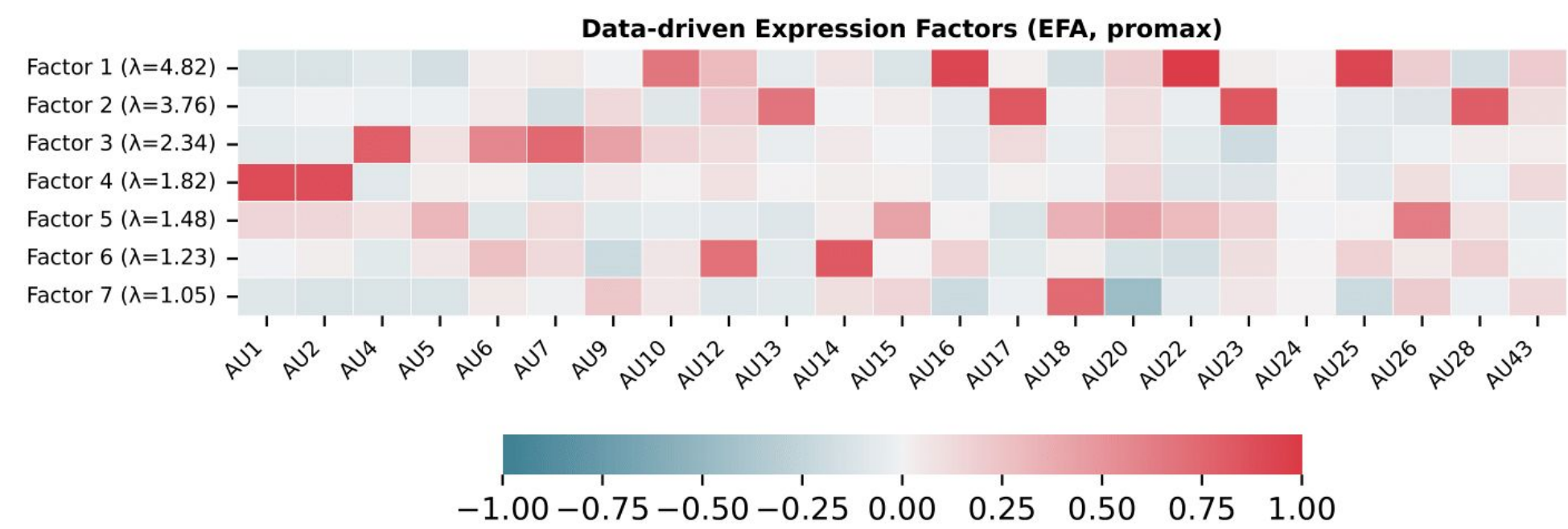
FCCS (Training):

$$w_{t,\tau} = \frac{\exp(s_{t,\tau}/\gamma)}{\sum_{\tau' \in \mathcal{L}} \exp(s_{t,\tau'}/\gamma)} \quad c_t^{\text{soft}} = \sum_{\tau \in \mathcal{L}} w_{t,\tau} \cdot s_{t,\tau}$$

Total Objective (Training):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_v + \lambda \mathcal{L}_{\text{soft-FCCS}} + \beta \mathcal{L}_{\text{jitter}}$$

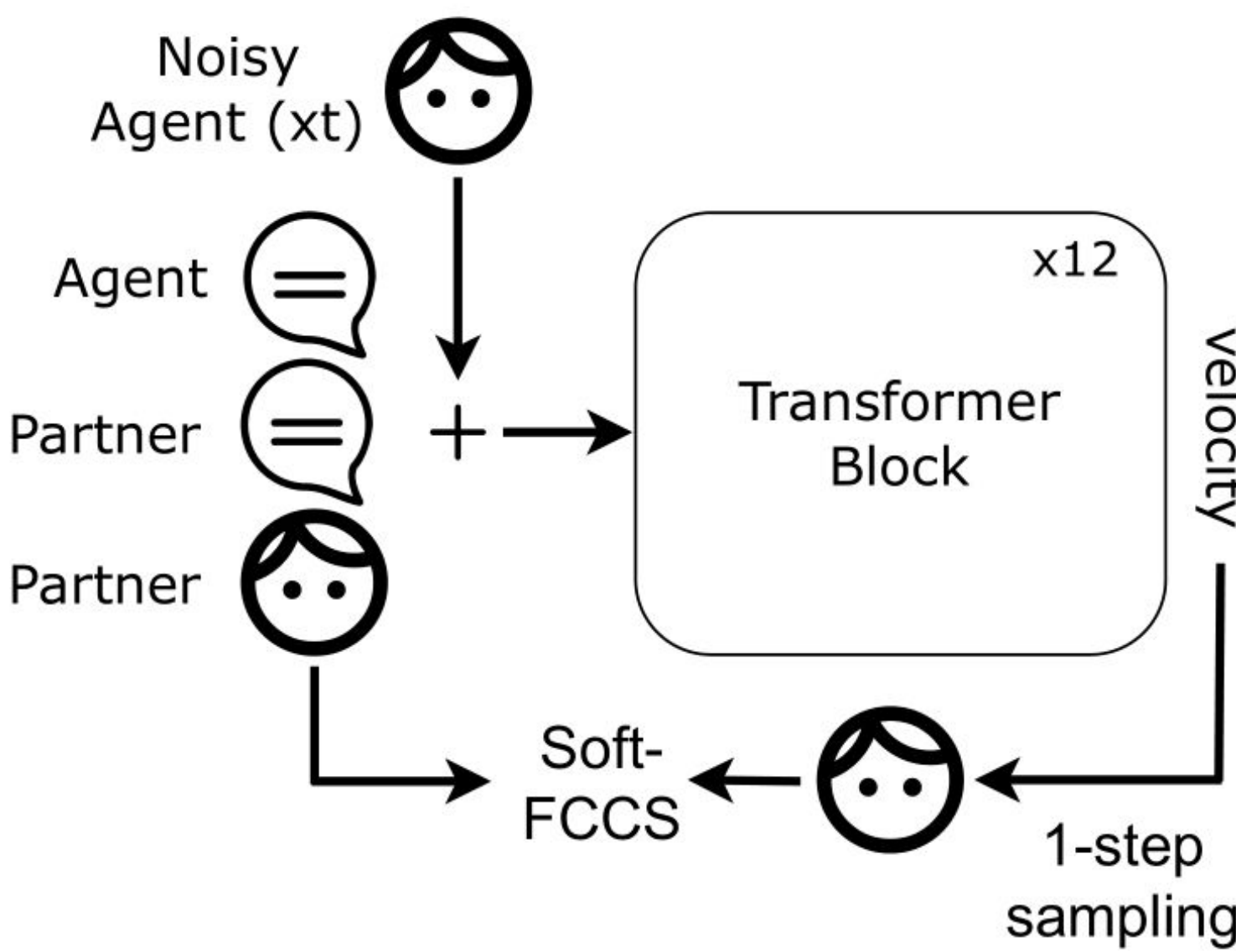
Facial Composite Discovery



Composite	Factor 6 (smiling)	Happiness	Factor 1 (mouth)	All others
Type	Data-Driven	Theory-Driven	Data-Driven	-
p_FDR	0.001	0.003	0.029	0.059–0.790

- Data-driven composites extracted via **Exploratory Factor Analysis (EFA)** on per-frame AU intensities
- Robust mimicry concentrates in **smiles (AU6+AU12)**, not arbitrary expressions
- Factor 1 (mouth) likely confounded by **shared audio**, not true visual mimicry

Modeling



- Modeling done on Seamless Interaction Dataset
- Learn a velocity field $v_{\theta}(x_t, t)$ that transports Gaussian noise to data; training uses **1-step clean estimate** from any noisy x_t
- Inference integrates the learned field via **100-step ODE solver** to generate motion sequences

Main Results

Target	Method	Non-Causal		Causal	
		MSE	Pearson r	MSE	Pearson r
Factor 1	Random	0.243 ± 0.076	0.240 ± 0.213	0.249 ± 0.074	0.249 ± 0.199
	Audio-Only	0.191 ± 0.052	0.410 ± 0.271	0.209 ± 0.071	0.335 ± 0.240
	Audio-Visual	0.185 ± 0.071	0.441 ± 0.233	0.198 ± 0.068	0.367 ± 0.251
	AV + soft-FCCS	0.177 ± 0.069	0.487 ± 0.213	0.187 ± 0.061	0.367 ± 0.219
Factor 6	Random	0.257 ± 0.092	0.240 ± 0.245	0.235 ± 0.099	0.207 ± 0.192
	Audio-Only	0.194 ± 0.050	0.356 ± 0.234	0.185 ± 0.086	0.380 ± 0.259
	Audio-Visual	0.161 ± 0.047	0.453 ± 0.298	0.146 ± 0.068	0.478 ± 0.237
	AV + soft-FCCS	0.123 ± 0.060	0.553 ± 0.199	0.132 ± 0.078	0.503 ± 0.239
Happiness	Random	0.243 ± 0.076	0.240 ± 0.265	0.215 ± 0.073	0.256 ± 0.220
	Audio-Only	0.201 ± 0.063	0.318 ± 0.271	0.186 ± 0.074	0.337 ± 0.231
	Audio-Visual	0.152 ± 0.071	0.413 ± 0.233	0.161 ± 0.071	0.408 ± 0.220
	AV + soft-FCCS	0.148 ± 0.078	0.445 ± 0.217	0.155 ± 0.078	0.447 ± 0.218

Model	β	MSE ↓	r ↑
Audio-Only	0	0.186 ± 0.074	0.337 ± 0.231
Audio-Visual	0	0.161 ± 0.071	0.408 ± 0.220
AV + soft-FCCS	0	0.155 ± 0.078	0.447 ± 0.218
Audio-Only	1	0.194 ± 0.072	0.325 ± 0.228
Audio-Visual	1	0.168 ± 0.073	0.391 ± 0.215
AV + soft-FCCS	1	0.159 ± 0.076	0.431 ± 0.213
Audio-Only	10	0.201 ± 0.078	0.298 ± 0.225
Audio-Visual	10	0.177 ± 0.077	0.362 ± 0.208
AV + soft-FCCS	10	0.167 ± 0.079	0.405 ± 0.207
Audio-Only	100	0.208 ± 0.083	0.271 ± 0.219
Audio-Visual	100	0.189 ± 0.082	0.328 ± 0.201
AV + soft-FCCS	100	0.178 ± 0.084	0.372 ± 0.203

- Visual conditioning improves fidelity and temporal agreement over audio-only (**r: 0.36 → 0.45**)
- soft-FCCS further sharpens timing, especially for smiles (**r: 0.45 → 0.55**, MSE -24%)
- Causal streaming **matches** non-causal look-ahead performance
- Jitter tradeoff: soft-FCCS **recovers expressivity** even under heavy smoothing (r: 0.34 → 0.40 at $\beta=100$)